

John Paul S

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Aspiring AI/ML Engineer from Prince Shri Venkateshwara Padmavathy Engineering College, driven by a passion for building intelligent systems at the intersection of deep learning, computer vision, and large language models.

TECHNICAL SKILLS

Languages Python, Java, HTML/CSS, JavaScript
AI / ML PyTorch, OpenCV, LoRA, Regression, Classification
LLMs GPT-2, Mistral, TinyLlama, DeepSeek, Gemini API, LLM Deployment

EXPERIENCE

RTRC Lab — National Taiwan University *Taipei, Taiwan | Jul – Nov 2025*
Research Intern (TEEP Scholar)

- Developed deep learning models for underwater image restoration under real-world aquatic conditions.
- Built a turbidity-aware restoration pipeline adaptive to varying water clarity levels.

Crion Technologies — IIT Madras Research Park *Chennai, India | Dec 2024 – Jul 2025*
AI / ML Intern

- Built end-to-end AI automation pipelines covering computer vision-based detection and OCR systems.
- Designed and deployed modular Python-based neural network models for production use.

Trinitysoft Solutions Pvt. Ltd. — United Kingdom *Remote | Jul – Oct 2025*
Architectural Fullstack Intern

- Fine-tuned and deployed LLMs (GPT-2, Mistral, TinyLlama) integrated into scalable web applications.
- Architected REST APIs connecting model inference to automated backend pipelines.

RESEARCH

APGR-Net: Adaptive Physics-Guided Restoration *Manuscript in preparation*
Physics constraints | Uncertainty modeling

- Applied Snell's law optical constraints with uncertainty estimation with several other real life constraints.

SU-CARE-FoB: Fog-Enabled Secure Medical IoT Framework *Paper accepted*
SDN routing | Urgency-factor prioritization

- Designed urgency-driven SDN routing for real-time triage data delivery in fog-enabled healthcare IoT networks.

PROJECTS

VisionPack: Intelligent Visual Quality Inspection Pipeline

- Curated and labeled a real-world industrial image dataset; trained YOLOv5 for high-accuracy defect detection.
- Applied SRGAN for image super-resolution to boost OCR fidelity; used EasyOCR and Tesseract for text extraction.
- Optimized inference latency for edge deployment on Microsoft Azure, ensuring real-time quality monitoring and reporting.

SensorSense: Reinforcement-Driven Predictive Maintenance Engine

- Ingested real-time sensor streams from industrial equipment and structured them for time-series analysis.
- Trained an LSTM model to identify degradation patterns; applied Q-Learning to generate prescriptive maintenance actions.
- Integrated a Flask-based monitoring dashboard to visualize Remaining Useful Life (RUL) and trigger automated alerts.

ACHIEVEMENTS & CERTIFICATIONS

Generative AI Certification — Nunnari Labs *2023*
Specialized in prompt engineering and fine-tuning large language models for creative and technical workflows.

Front-End Developer Certification — Greens Technology *2024*
Mastered responsive web design and modern JavaScript frameworks to build high-performance user interfaces.

TEEP International Internship — National Taiwan University *2025*
Selected for a prestigious global research fellowship focused on advanced computer vision and image restoration.

NCC National Level Camp Achiever *2025*
Attended National level NCC Camp for outstanding leadership and communication skills.